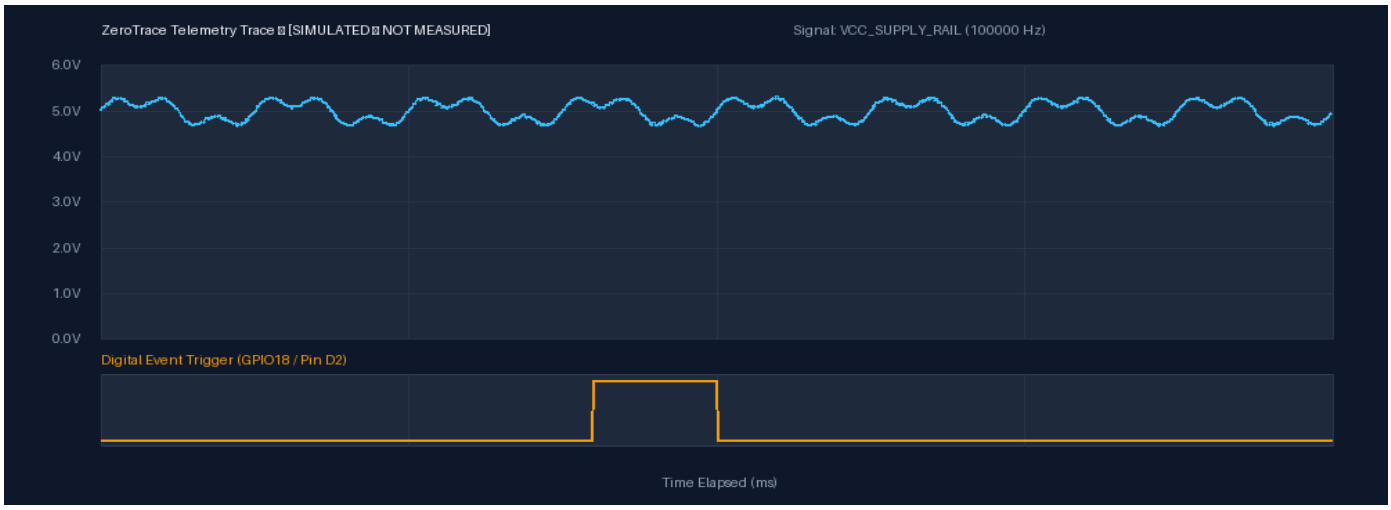
ZEROTRACE — DEFENSIVE TELEMETRY & AUDIT REPORT

Run Identifier: **CMP-04a161** | Generated: 2026-10-29 01:04:38 UTC | Target Layer: **Layer 1**

Scenario / Campaign	Full 40-Problem Comprehensive Test Campaign	**Data Provenance Mode**	**SIMULATED — NOT MEASURED**
Executed Tests	13	**Successful Mitigations**	13
Detection Rate	100.0%	**Containment Rate**	100.0%
Mean Detection Latency	1.2 ms	**Mean Defense Latency**	2.8 ms

SYNCHRONIZED TIME-SERIES WAVEFORM & GLITCH TRANSIENT



RECORDED EVENT TIMELINE & DEFENSIVE DECISIONS

Test ID	Event Type	Observed State	Defense Action	Risk	Outcome
test_l1_01_untrust	UNTRUSTED_ISSUER_REJECTE	UNAUTHORIZED_ISSUER	REJECT_UNTRUSTED_PACKAGE	MEDIUM	PASS
test_l1_01_untrust	PACKAGE_VERIFICATION_SUC	TRUSTED_ORIGIN	ACCEPT_PACKAGE_FROM_ROOT_A	LOW	PASS
test_l1_02_payload	HASH_MISMATCH_TAMPER_DET	SHA256_MISMATCH	ABORT_AND_PURGE_STAGING_BU	HIGH	PASS
test_l1_03_signatu	SIGNATURE_INVALID_REJECT	FORGED_OR_FAULTED_SIGNAT	HALT_SECURE_BOOT_CHAIN	HIGH	PASS
test_l1_04_replay_	NONCE_FRESH_ACCEPTED	FRESH_TRANSACTION	RECORD_NONCE_AND_PROCEED	LOW	PASS
test_l1_04_replay_	REPLAY_DETECTED_REJECTED	NONCE_REUSED_OR_EXPIRED	DISCARD_REPLAY_PACKET	MEDIUM	PASS

test_l1_05_anti_ro	DOWNGRADE_BLOCKED	VERSION_100_LESS_THAN_12	ENFORCE_ANTI_ROLLBACK_POLI	HIGH	PASS
test_l1_05_anti_ro	VERSION_ACCEPTED_COUNTER	MONOTONIC_INCREASE	UPDATE_SECURE_NVM_VERSION	LOW	PASS
test_l1_06_malform	MANIFEST_SCHEMA_INVALID	MISSING_MANDATORY_FIELDS	ABORT_HEADER_PARSER	MEDIUM	PASS
test_l1_07_hardwar	TARGET_MISMATCH_REJECTED	INCOMPATIBLE_TARGET_WRON	BLOCK_FLASH_PROGRAMMING	HIGH	PASS

SCIENTIFIC TRACEABILITY & CITATIONS

Research Traceability References:

• **[Wiersma-2017]**: *Safety != Security: On the resilience of ASIL-D certified microcontrollers against fault injection attacks* (FDTC 2017). Supports dual-counter loop checking, lockstep delay, and JTAG DPA profiling.
• **[Fanjas-2024]**: *PoP DRAM: A new EMFI approach based on EM-induced glitches on SoC* (FDTC 2024). Supports F1-F9 instruction skip models and capacitor EM pulse injection.
• **[Fanjas-2023]**: *Exploration of SoC Secure-Boot Vulnerability to Fault-Injection by Side-Channel Analysis* (IEEE DFT 2023). Supports SSBL certificate chain verification and memcmp hash SCA.
• **[Luo-2022]**: *On Security of TrustZone-M-Based IoT Systems* (IEEE IoTJ 2022). Supports DAL tiering, RXN/DXN non-executable policies, anti-rollback secure flash, and ATECC608A key isolation.